CSM202: Statistical Inference

Sampling Distributions, Confidence Intervals, and Hypothesis Testing

Lecture 08

Kezang Yuden(PhD), Mani Wangchuk, and Chimi Dendup

Gyalpozhing College of Information Technology

chimidendup.gcit@rub.edu.bt
+975 - 77365024

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Learning Outcomes

By the end of this lecture, students should be able to:

- ☐ distinguish parameters, estimators, estimates, and sampling distributions;
- ☐ derive the mean and variance of the sample mean;
- ☐ compute standard errors for sample means and proportions;
- ☐ explain and apply the central limit theorem;
- ☐ construct confidence intervals for a population mean when σ is known;
- ☐ construct Student's t confidence intervals when σ is unknown;
- ☐ construct confidence intervals for a population proportion;
- ☐ interpret confidence level correctly;
- ☐ determine a required sample size for a target margin of error;
- ☐ state and formulate null and alternative hypotheses;
- ☐ distinguish one-tailed and two-tailed tests;
- ☐ define Type I and Type II errors;
- ☐ explain significance level and statistical power;

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Definition

A **population** is the complete set of units of interest. A **sample** is a subset observed from the population. A **parameter** is a numerical feature of the population, and a **statistic** is a numerical feature computed from a sample.

- Parameter examples: population mean μ, population proportion p.
- Statistic examples: sample mean $\bar{x}$, sample proportion $\hat{p}$.
- We usually know the statistic and use it to estimate the parameter.

Notation for this Lecture

Symbol	Meaning
$X_1, X_2, \ldots, X_n$	random sample of size n
μ	unknown population mean
σ	population standard deviation
$\bar{X} = \frac{1}{n}\sum X_i$	sample mean as a random variable
$\bar{x}$	observed value of the sample mean
s	sample standard deviation
p	unknown population proportion
$\hat{p}$	sample proportion

Key Idea

Uppercase notation usually describes random quantities before data are observed; lowercase describes observed values.

Estimator and Estimate

Definition

An **estimator** is a rule or formula for estimating a parameter. An **estimate** is the numerical value produced after applying the estimator to data.

- Estimator: $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$
- Estimate: if observed values have mean 182, then $\bar{x} = 182$.
- Estimators vary from sample to sample; estimates are fixed numbers.

Bias, Consistency, Efficiency [1]

Definition

An estimator T of parameter θ is **unbiased** if $E(T) = \theta$. It is **consistent** if it gets closer to θ as n grows. It is more **efficient** if it has smaller variance among comparable estimators.

- Bias: average error over repeated samples.
- Consistency: long-run improvement with more data.
- Efficiency: less sampling variability.

Sampling Distribution

Definition

The **sampling distribution** of a statistic is the probability distribution of that statistic under repeated random sampling from the population.

- It describes uncertainty caused by sampling.
- It is not the distribution of raw data.
- It is the distribution of a statistic such as $\bar{X}$ or $\hat{p}$.

Mean and Variance of the Sample Mean

Assume $X_1, X_2, \ldots, X_n$ are independent observations with common mean μ and variance σ^2.

$$E[\bar{X}] = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)$$
$$= \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n}(n\mu) = \mu.$$

Key Idea

$\bar{X}$ is unbiased for μ.

Variance of the Sample Mean

Using independence:

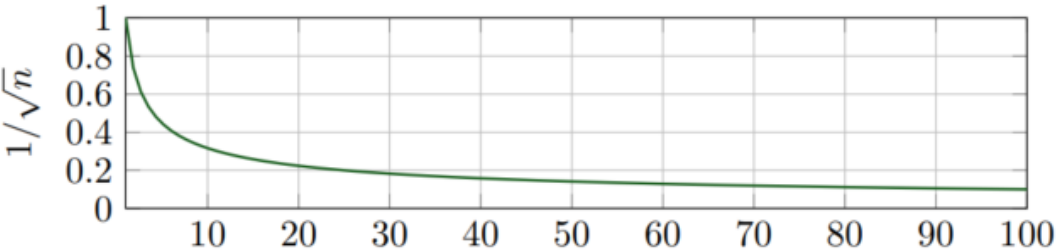
$$\begin{aligned} Var(\bar{X}) &= Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\ &= \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) \\ &= \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}. \end{aligned}$$

Key Idea

The **standard error** of $\bar{X}$ is $\sigma_{\bar{X}} = \sigma/\sqrt{n}$.

Standard Error: Intuition

- Standard deviation describes spread of individual observations.
- Standard error describes spread of an estimator over repeated samples.
- Increasing n reduces standard error by a square-root law.



Central Limit Theorem(CLT)

Definition

The central limit theorem states that even if a population distribution is strongly non-normal, its sampling distribution of means will be approximately normal for large sample sizes (over 30). The central limit theorem makes it possible to use probabilities associated with the normal curve to answer questions about the means of sufficiently large samples.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

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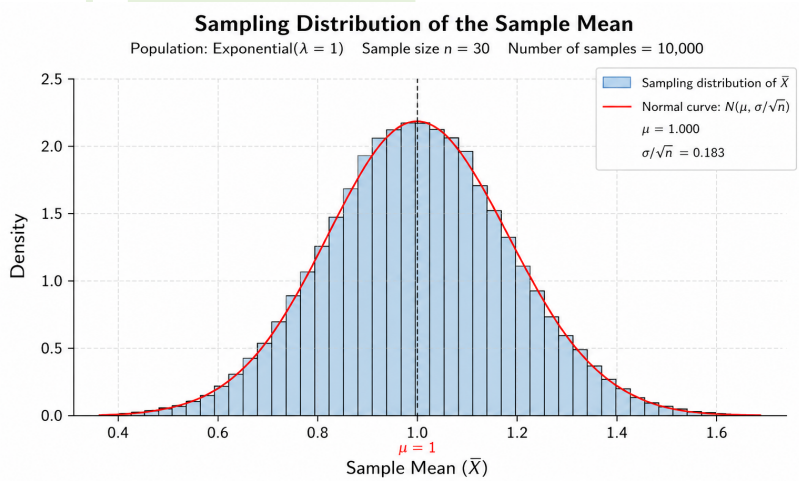
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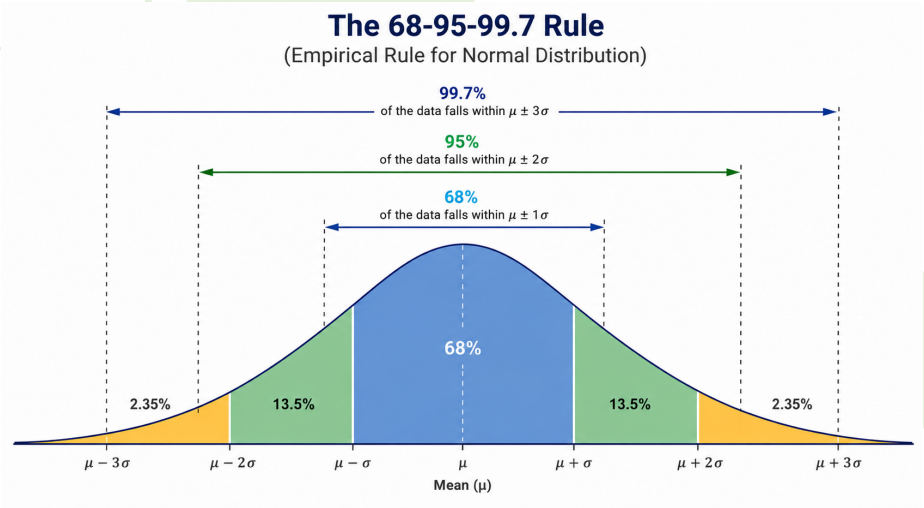
The CLT is the bridge from probability distributions to statistical inference.

CLT Visual Explanation



The sampling distribution of the sample mean is approximately normal with mean μ and standard error $\sigma/\sqrt{n}$.

68-95-99 Rule for Normal Distribution



Confidence Interval

Definition

A **confidence interval** is an interval computed from sample data using a method that captures the true parameter in a specified proportion of repeated samples.

Key Idea

Most confidence intervals have the structure

$$\text{estimate} \pm \text{critical value} \times \text{standard error}$$

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Common Mistake

A 95% confidence interval does not mean there is a 95% probability that this fixed interval contains the fixed parameter. The 95% describes the long-run method.

Known Variance: z Confidence Interval for a Mean

Assume σ is known and $\bar{X}$ is approximately normal.

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0,1)$$

For 95% interval, use $Z_{0.975} = 1.96$:

$$\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

- $\bar{x}$: observed sample mean.
- $\sigma/\sqrt{n}$: standard error.
- 1.96: middle 95% cutoff of standard normal distribution.

Deriving the z Interval

Start from the central 95% normal probability:

$$P\left(-1.96 \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq 1.96\right) \approx 0.95$$

Rearrange the middle inequality:

$$-1.96\frac{\sigma}{\sqrt{n}} \leq \bar{X}-\mu \leq 1.96\frac{\sigma}{\sqrt{n}}$$

$$\boxed{\bar{X}-1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}+1.96\frac{\sigma}{\sqrt{n}}}$$

Replace $\bar{X}$ by observed $\bar{x}$ after collecting data.

Common critical values:

Confidence Level	$Z_{\alpha/2}$
90%	1.645
95%	1.96
99%	2.576

Solved Examples

Example

A university administrator wants to estimate the average amount of time students spend using the computer laboratory each week. A random sample of 64 students gives a mean laboratory usage time of 18.5 hours per week. The population standard deviation is known to be 4.8 hours.

Construct a 95% confidence interval for the true mean weekly laboratory usage time of all students. Interpret the result.

Solution: Given: $\bar{x} = 18.5, \sigma = 4.8, n = 64, Z_{0.025} = 1.96$
The 95% confidence interval is

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Substituting the values,

$$\begin{aligned} 18.5 \pm 1.96\left(\frac{4.8}{\sqrt{64}}\right) &= 18.5 \pm 1.96\left(\frac{4.8}{8}\right) \\ &= 18.5 \pm 1.96(0.6) \\ &= 18.5 \pm 1.176 \\ &= (17.324, 19.676). \end{aligned}$$

Hence, the 95% confidence interval is $(17.324, 19.676)$.
Interpretation: We are 95% confident that the true mean weekly computer laboratory usage time of all students lies between 17.32 and 19.68 hours.

Unknown Variance: Why t?

- In most real studies, σ is unknown.
- Replacing σ by s adds extra uncertainty.
- Student's t distribution accounts for this extra uncertainty.

Definition

For a normal population, or a large sample, the statistic

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

follows approximately a t distribution with $n - 1$ degrees of freedom.

t Confidence Interval for a Mean

When σ is unknown:

$$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

Part	Meaning
α	total non-coverage probability, e.g. 0.05 for 95% confidence
$t_{\alpha/2,n-1}$	positive critical value from t distribution
$n-1$	degrees of freedom from estimating μ using $\bar{x}$
$s/\sqrt{n}$	estimated standard error

Solved Examples

Example

A lecturer wants to estimate the average number of hours students spend studying each week. A random sample of 16 students has a mean study time of 22.5 hours and a sample standard deviation of 4.8 hours.

Construct a 95% confidence interval for the true mean weekly study time of all students, assuming the study times are approximately normally distributed.

Solution: Given, $\bar{x} = 22.5, s = 4.8, n = 16, df = 15$.
From the Student's t-table, $t_{0.025,15} = 2.131$.
The 95% confidence interval for the population mean is

$$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}.$$

Substituting the given values,

$$\begin{aligned} 22.5 \pm 2.131\left(\frac{4.8}{\sqrt{16}}\right) &= 22.5 \pm 2.131\left(\frac{4.8}{4}\right) \\ &= 22.5 \pm 2.557 \\ &= (19.943,\ 25.057). \end{aligned}$$

Hence, the 95% confidence interval is $\boxed{(19.94,\ 25.06)}$.

Interpretation: We are 95% confident that the true mean weekly study time of all students lies between **19.94** and **25.06** hours.

Try Yourself

1. A machine fills bottles with juice. A sample of 100 bottles gives a mean fill of 500 mL. Assume that the standard deviation is 12 mL. Find the 99 % confidence interval for the true mean fill.
2. A researcher measures 20 reaction times with a mean of 0.85 seconds and a standard deviation of 0.12 seconds. Construct a 98 % confidence interval for the mean reaction time.

What Is a Statistical Hypothesis?

Definition

A **statistical hypothesis** is a claim about an unknown population parameter, such as a population mean μ, a population proportion p, or a difference of means $\mu_1 - \mu_2$.

- Example parameter: $\mu =$ true mean response time after deployment.
- Example claim: $\mu < 200ms$.
- Data are used to judge whether the claim is plausible.

Null and Alternative Hypotheses

Alternative hypothesis (H_a)

Denoted by the term H_a or H_1, this hypothesis is what one would believe if the null hypothesis does not hold. (**e.g.** students in situation A will perform better than students in situation B)

Null and Alternative Hypotheses

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Null hypothesis (H_0)

This is the hypothesis that needs to be tested. It is based on the principle that there is no change from the status quo. (**e.g.** *there will be no difference in the performance of students in situations A and B*)

- H_0 usually contains equality.
- H_a can be one-sided or two-sided.
- We test evidence against H_0.

One-Sided and Two-Sided Alternatives

Question	Alternative	Example
Has mean decreased?	$H_a : \mu < \mu_0$	New cache lowers latency
Has mean increased?	$H_a : \mu > \mu_0$	Defect rate rose after release
Has mean changed?	$H_a : \mu \neq \mu_0$	New compiler changes runtime

One-Sided and Two-Sided Alternatives

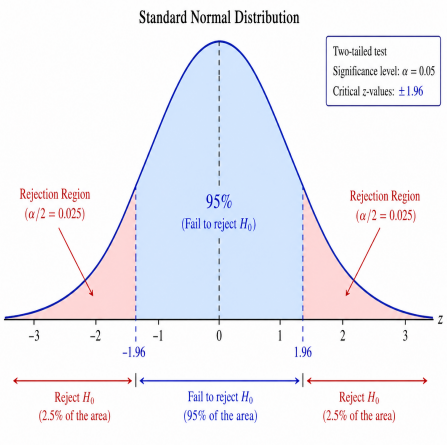
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Common Mistake

Do not choose one-sided or two-sided after looking at the sample. The direction must come from the research question before testing.

Rejection Region / Levels of Significance (α)

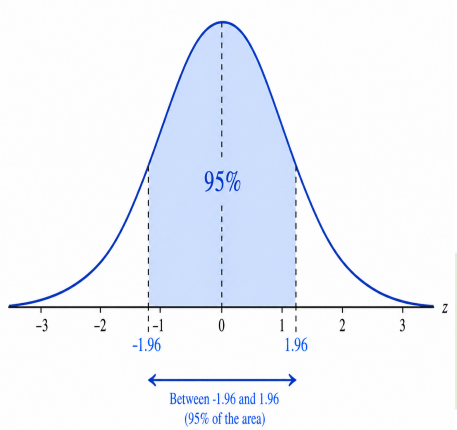
- Used to indicate the chance that we are wrong in rejecting the null hypothesis.
- Also called the level of probability or *p* level.
- We have to ensure that the probability of rejecting a true null hypothesis is very small.
- The frequently used α values are $\alpha = 0.05(95\%)$, $\alpha = 0.01(99\%)$.



Level of Confidence (c)

- Confidence level is a measure of the coverage probability of a confidence interval is called confidence level. The relationship between level of confidence and level of significance is given below:

$$c = 1 - \alpha$$



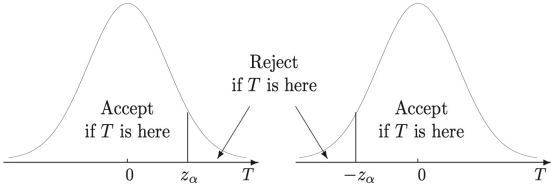
The p - value

Definition

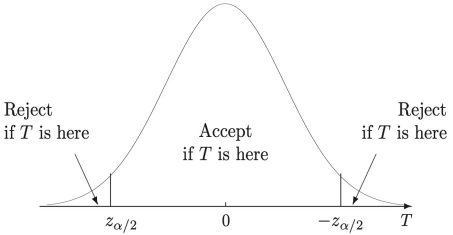
The **p-value** is the probability, assuming H_0 is true, of obtaining a test statistic at least as extreme as the one observed.

- For a two-sided z test: $p_{value} = 2P(Z \geq |z_{obs}|)$
- For a right-tailed test: $p_{value} = P(Z \geq z_{obs})$
- For a left-tailed test: $p_{value} = P(Z \leq z_{obs})$
- Decision rule: $p_{value} \leq \alpha \implies$ reject H_0.

Acceptance and Rejection Region for Z-text



(a) *Right-tail Z-test* (b) *Left-tail Z-test*



(c) *Two-sided Z-test*

Step-by-Step Testing Procedure

1. State the null and alternate hypothesis
2. Fix the level of significance and obtain the critical value of the test statistic
3. Select the appropriate test: Choose the test based on the following parameters:
   - number of samples
   - whether the population is normally distributed
   - statistic being tested
   - sample size
   - whether the population standard deviation is known.
4. Obtain the relevant test statistic (z statistic/t statistic/chi-square statistic/f statistic) or the p-value
5. Compare the critical test statistic with the calculated test static or compare the p-value with 0.05.
6. Draw an inference based on the preceding comparison.

Type I and Type II Errors

	Accept Null	*Reject Null*
Null is true	Correct decision: there is no difference	Type I error
Null is false	Type II error	Correct decision: there are differences

- In two of the four cases, the test result in a correct decision (accept a true hypothesis or reject false hypothesis).
- **Type I error** - when we reject a true hypothesis
- **Type II error** - when we accept a false hypothesis
- Probability of type I error is the significance level of a test ($\alpha = P\{$ reject $H_0 \mid H_0 istrue\}$).

Note

A type I error is considered more dangerous. Making a type I error can be compared with convicting an innocent defendant or sending a patient to surgery when he does not need one.

Example

The school board claims that at least 60% of students bring a phone to school. A teacher believes this number is too high and randomly samples 25 students to test at a level of significance of 0.02. Write H_0 and H_a.

Solution :

$$H_0 : p \geq 0.60$$
$$H_a : p < 0.60$$
$$\alpha = 0.02 (\text{Level of significance})$$
$$c = 1 - \alpha = 0.98 (\text{Confidence level})$$

Solved Examples

Example

A machine is producing perfume bottles with the long term average of 150 *cc* and the standard deviation of 2 *cc*. Four bottles were picked and the average volume was found to be 153.8 *cc*. Has the average volume changed from 150 *cc*? Test with 95% confidence level.

Solution:

First, let us state null and alternate hypothesis:

$$H_0 : \mu = 150 \ cc$$
$$H_a : \mu \neq 150 \ cc$$
$$\alpha = 1 - 0.95 = 0.05 \text{(Significance Level)}$$

The Z calculated will be calculated using the formula given below;

$$Z_{Calculated} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$
$$= \frac{153.8 - 150}{2/\sqrt{4}} = 3.8$$

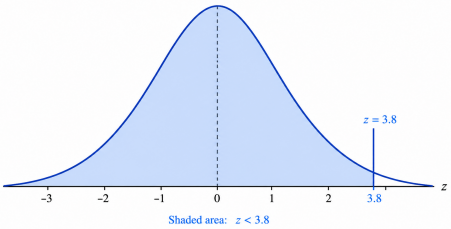
Now, the Z critical be found from the Z - table. But, since the volume can be less than or more than the null hypothesis. Here, we shall use **two tail** test. Significance level will be checked for $(\alpha/2 = 0.025)$.

$$\therefore Z_{Critical} = 1.96$$

Our calculated Z value is in the rejection zone. As a result, **we reject the null hypothesis**.

Using p-value

Since this is a two-tailed test, the p-value is: $p\text{-value} = 2P(Z > 3.8)$
Using standard normal tables:
$P(Z > 3.8) \approx 0.00007$
So, $p\text{-value} = 2 \times 0.00007 \approx 0.00014$.
Since $p\text{-value} = 0.00014 < 0.05$, we reject H_0, confirming that the mean volume has significantly changed from 150 cc.



Conclusion

At the 5% significance level, we reject the null hypothesis H_0. There is sufficient statistical evidence to conclude that the true mean volume of perfume bottles differs from 150 cc.

Solved Example

If an unauthorized person accesses a computer account with the correct username and password (stolen or cracked), can this intrusion be detected? Recently, a number of methods have been proposed to detect such unauthorized use. The time between keystrokes, the time a key is depressed, the frequency of various keywords are measured and compared with the account owner. If there are noticeable differences, an intruder is detected.
The following times between keystrokes were recorded when a user typed the username and password:

$$.46, .38, .31, .24, .20, .31, .34, .42, .09, .18, .46, .21 \textit{seconds}$$

1. As the first step in detecting if this is an intrusion, construct a 90 % confidence interval for the mean time between keystrokes assuming Normal distribution of these times.
2. A long-time authorized user of the account makes 0.2 seconds between keystrokes. One day, the data above are recorded as someone typed the correct username and password. At a 1 % level of significance, is this an evidence of an unauthorized attempt?

Solved Example

Solution

1. *The sample size is* $n = 12$*, the sample mean time is* $\bar{X} = 0.3$ *sec, and the sample standard deviation is* $s = 0.1183$*. The critical value of the t-distribution with* $n - 1 = 11$ *degrees of freedom is* $t_{0.05,\,11} = 1.796$ *(upper-tail area* 0.05*, i.e.* $t_{\alpha/2}$ *with* $\alpha = 0.10$*). Then, the* 90% *confidence interval for the mean time is*

$$0.3 \pm (1.796)\frac{0.1183}{\sqrt{12}} = 0.3 \pm 0.0613 = [0.2387,\ 0.3613]$$

Solution

1. *2. Let us test*

$$H_0 : \mu = 0.2 \text{ vs } H_A : \mu \neq 0.2$$

at a significance level $\alpha = 0.01$. *We have sample statistics* $n = 12, \bar{X} = 0.3$ *and* $s = 0.1183$. *Compute the T-statistic,*

$$t = \frac{\bar{X} - 0.2}{s\sqrt{n}} = \frac{0.3 - 0.2}{0.1183\sqrt{12}} = 5.8565.$$

The acceptance region is $[3.106, 3.106]$ *(we used T-distribution with* $12 - 1 = 11$ *degrees of freedom and* $\alpha/2 = 0.005$ *because of the two-sided alternative), and it does not include our test statistic. Therefore, we reject the null hypothesis and conclude that there is a significant evidence of an unauthorized use of that account.*

Try Yourself

1. The number of concurrent users for some internet service provider has always averaged 5000 with a standard deviation of 800. After an equipment upgrade, the average number of users at 100 randomly selected moments of time is 5200. Does it indicate, at a 5 % level of significance, that the mean number of concurrent users has increased? Assume that the standard deviation of the number of concurrent users has not changed.

References

[1] Gayathri Rajagopalan. A python data analyst's toolkit: Learn python and python-based libraries with applications in data analysis and statistics. 2021.